



What is the battery of the future made of?

Researchers at the Swiss Federal Laboratories are investigating new materials for next-gen batteries, ranging from fast-charging options for vehicles to cost-effective solutions for stationary storage, each with bespoke materials



Kostiantyn Kravchyk studies whether the unwanted growth of dendrites can be reduced with pressure. The work on the novel batteries is carried out in a protective atmosphere (Credit: Empa)

and innovative manufacturing techniques. The team has enhanced the solid electrolyte lithium lanthanum zirconium oxide (LLZO) for its ionic conductivity and chemical stability, creating a bilayer membrane where one layer stores lithium to maximise contact and minimise current density, and the other acts as a dendrite growth barrier. This approach merges effectiveness with cost-efficiency in manufacturing. Tackling the high cost of lithium-ion batteries, the team also explored iron as a more affordable cathode material, overcoming its low electrical and lithium ion conductivity by innovating a cost-effective method to enhance iron (III) hydroxychloride with a pyrochlore crystalline structure for effective lithium ion conduction, presenting an economical solution for battery cathodes.

Averting catastrophic breakdown in lithium-ion batteries

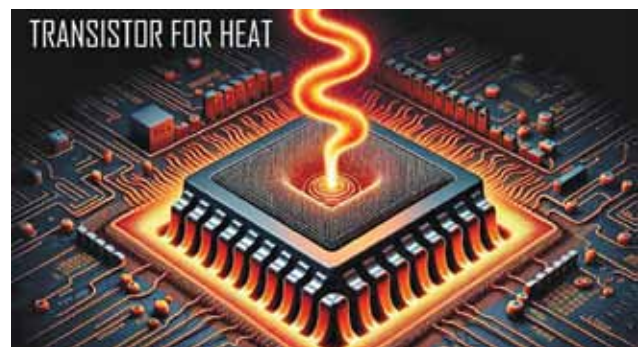
Scientists from the Australian Nuclear Science and Technology Organisation (ANSTO) have used neutron scattering methods to investigate detrimental lithium formations in rechargeable lithium-ion batteries (LIBs). Small-angle and ultra-small-angle neutron scattering techniques were employed to identify the size and shape of lithium structures without dismantling the battery. This approach, more efficient than methods like X-ray imaging or microscopy, allows for the observation of lithium deposition processes during operation, providing valuable insights into battery performance.

World's first dynamic resource allocation technology

Fujitsu unveiled the world's inaugural technology for dynamic resource allocation between CPUs and GPUs in real-time. This prioritises the efficient execution of GPU-intensive programs to address the global GPU shortage, driven by demand in generative AI, deep learning, and other fields, while maximising existing computational resources. The company has introduced a parallel processing technology in high-performance computing (HPC) systems, allowing seamless real-time transitions between programs and linking numerous computers for large computations. This technology enables the immediate start of resource-intensive applications without waiting for ongoing programs and will be integrated into a developing computer workload broker that allows AI to autonomously select computing resources based on various factors. The company has also created a technology that differentiates between GPU and CPU suitable programs during concurrent execution, employing predictive acceleration rate assessment and real-time GPU allocation. These advancements aim to provide accessible, cost-effective, and high-performance computing resources, contributing to a society equipped with advanced computational capabilities.

A solid-state thermal transistor for precise heat control

Researchers from the University of California, Los Angeles (UCLA), have unveiled a solid-state thermal transistor leveraging an electric field for precise control of heat



Thermal transistor for precise control of heat (Credit: <https://techxplore.com>)

movement within a semiconductor device. Overcoming limitations with a solid-state design and field effect, this innovation enables modulation of thermal conductivity at the atomic level for heat flux switching. With switching speeds over 1 megahertz and high tunability in thermal conductance, it holds promise for sustainable energy applications in chip manufacturing and performance.